CSD19533Q5A 100 V N-Channel NexFET™ Power MOSFET

1 Features

- Ultra-Low Q_g and Q_{gd}
- Low Thermal Resistance
- Avalanche Rated
- Pb-Free Terminal Plating
- RoHS Compliant
- Halogen Free
- SON 5-mm × 6-mm Plastic Package

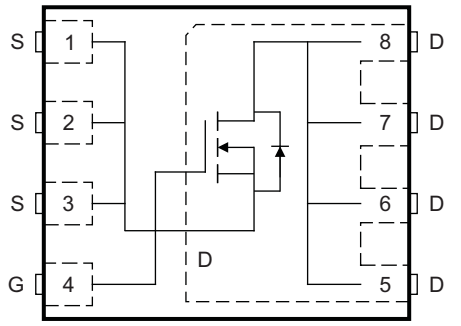
2 Applications

- Primary Side Telecom
- Secondary Side Synchronous Rectifier
- Motor Control

3 Description

This 100 V, 7.8 mΩ, SON 5 mm × 6 mm NexFET™ power MOSFET is designed to minimize losses in power conversion applications.

Top View



P0093-01

Product Summary

T_A = 25°C		TYPICAL VALUE		UNIT
V_{DS}	Drain-to-Source Voltage	100		V
Q_g	Gate Charge Total (10 V)	27		nC
Q_{gd}	Gate Charge Gate to Drain	4.9		nC
$R_{DS(on)}$	Drain-to-Source On Resistance	V_{GS} = 6 V	8.7	mΩ
		V_{GS} = 10 V	7.8	mΩ
$V_{GS(th)}$	Threshold Voltage	2.8		V

Ordering Information(1)

Device	Media	Qty	Package	Ship
CSD19533Q5A	13-Inch Reel	2500	SON 5 x 6 mm Plastic Package	Tape and Reel
CSD19533Q5AT	7-Inch Reel	250		

(1) For all available packages, see the orderable addendum at the end of the data sheet.

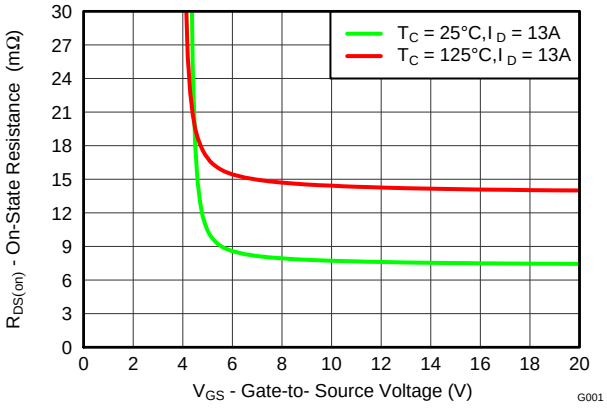
Absolute Maximum Ratings

T_A = 25°C		VALUE	UNIT
V_{DS}	Drain-to-Source Voltage	100	V
V_{GS}	Gate-to-Source Voltage	±20	V
I_D	Continuous Drain Current (Package limited)	100	A
	Continuous Drain Current (Silicon limited), T_C = 25°C	75	
	Continuous Drain Current, T_A = 25 °C(1)	13	
I_{DM}	Pulsed Drain Current, T_A = 25 °C(2)	231	A
P_D	Power Dissipation(1)	3.2	W
	Power Dissipation, T_C = 25°C	96	
T_J, T_{stg}	Operating Junction and Storage Temperature Range	–55 to 150	°C
E_{AS}	Avalanche Energy, single pulse I_D = 46 A, L = 0.1 mH, R_G = 25 Ω	106	mJ

(1) Typical $R_{\theta JA}$ = 40 °C/W on a 1-inch^2, 2-oz. Cu pad on a 0.06-inch thick FR4 PCB.

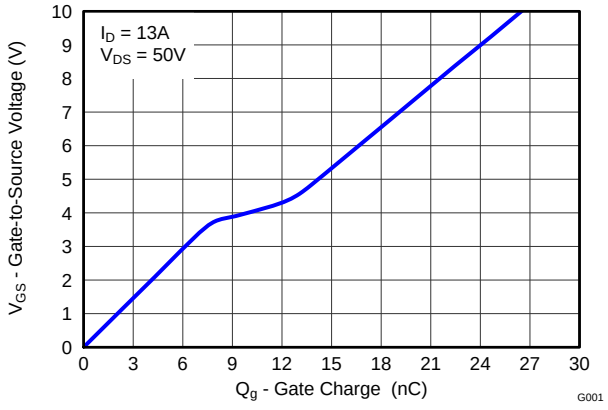
(2) Max $R_{\theta JC}$ = 1.3°C/W, pulse duration ≤100 µs, duty cycle ≤1%

$R_{DS(on)}$ vs V_{GS}



G001

Gate Charge



G001

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

Changes from Original (December 2013) to Revision A **Page**

- Added small reel order number ... 1
- Increased pulsed drain current to 231A ... 1
- Added line for max power dissipation with case temperature held to 25°C ... 1
- Updated the pulsed drain current conditions ... 1
- Fixed y-axis on Figure 1 to state that it is a normalized $R_{\theta JC}$ curve ... 4
- Updated the safe operating area in Figure 10 ... 6

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5 Specifications

5.1 Electrical Characteristics

(T_A = 25°C unless otherwise stated)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
STATIC CHARACTERISTICS						
BV_{DSS}	Drain-to-Source Voltage	V_{GS} = 0 V, I_D = 250 µA	100			V
I_{DSS}	Drain-to-Source Leakage Current	V_{GS} = 0 V, V_{DS} = 80 V			1	µA
I_{GSS}	Gate-to-Source Leakage Current	V_{DS} = 0 V, V_{GS} = 20 V			100	nA
$V_{GS(th)}$	Gate-to-Source Threshold Voltage	V_{DS} = V_{GS}, I_D = 250 µA	2.2	2.8	3.4	V
$R_{DS(on)}$	Drain-to-Source On Resistance	V_{GS} = 6 V, I_D = 13 A		8.7	11.1	mΩ
		V_{GS} = 10 V, I_D = 13 A		7.8	9.4	mΩ
g_{fs}	Transconductance	V_{DS} = 10 V, I_D = 13 A		63		S
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	V_{GS} = 0 V, V_{DS} = 50 V, f = 1 MHz		2050	2670	pF
C_{oss}	Output Capacitance			395	514	pF
C_{rss}	Reverse Transfer Capacitance			9.6	12.5	pF
R_G	Series Gate Resistance			1.2	2.4	Ω
Q_g	Gate Charge Total (10 V)	V_{DS} = 50 V, I_D = 13 A		27	35	nC
Q_{gd}	Gate Charge Gate to Drain			4.9		nC
Q_{gs}	Gate Charge Gate to Source			7.9		nC
$Q_{g(th)}$	Gate Charge at V_{th}			5.7		nC
Q_{oss}	Output Charge	V_{DS} = 50 V, V_{GS} = 0 V		75		nC
$t_{d(on)}$	Turn On Delay Time	V_{DS} = 50 V, V_{GS} = 10 V, I_{DS} = 13 A, R_G = 0 Ω		6		ns
t_r	Rise Time			6		ns
$t_{d(off)}$	Turn Off Delay Time			16		ns
t_f	Fall Time			5		ns
DIODE CHARACTERISTICS						
V_{SD}	Diode Forward Voltage	I_{SD} = 13 A, V_{GS} = 0 V		0.8	1.0	V
Q_{rr}	Reverse Recovery Charge	V_{DS}= 50 V, I_F = 13 A, di/dt = 300 A/µs		163		nC
t_{rr}	Reverse Recovery Time			62		ns

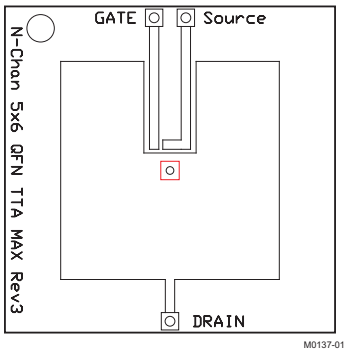
5.2 Thermal Information

(T_A = 25°C unless otherwise stated)

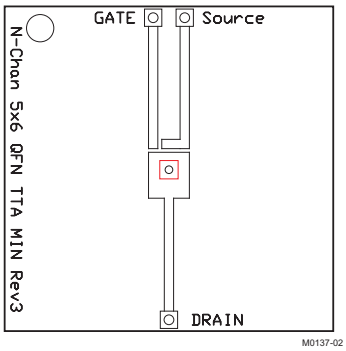
THERMAL METRIC		MIN	TYP	MAX	UNIT
$R_{\theta JC}$	Junction-to-Case Thermal Resistance(1)			1.3	°C/W
$R_{\theta JA}$	Junction-to-Ambient Thermal Resistance(1)(2)			50	

(1) $R_{\theta JC}$ is determined with the device mounted on a 1-inch^2 (6.45-cm^2), 2-oz. (0.071-mm thick) Cu pad on a 1.5-inches × 1.5-inches (3.81-cm × 3.81-cm), 0.06-inch (1.52-mm) thick FR4 PCB. $R_{\theta JC}$ is specified by design, whereas $R_{\theta JA}$ is determined by the user's board design.
(2) Device mounted on FR4 material with 1-inch^2 (6.45-cm^2), 2-oz. (0.071-mm thick) Cu.

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Max $R_{\theta JA}$ = 50°C/W when mounted on 1 inch^2 (6.45 cm^2) of 2-oz. (0.071-mm thick) Cu.



Max $R_{\theta JA}$ = 115°C/W when mounted on a minimum pad area of 2-oz. (0.071-mm thick) Cu.

5.3 Typical MOSFET Characteristics

(T_A = 25°C unless otherwise stated)

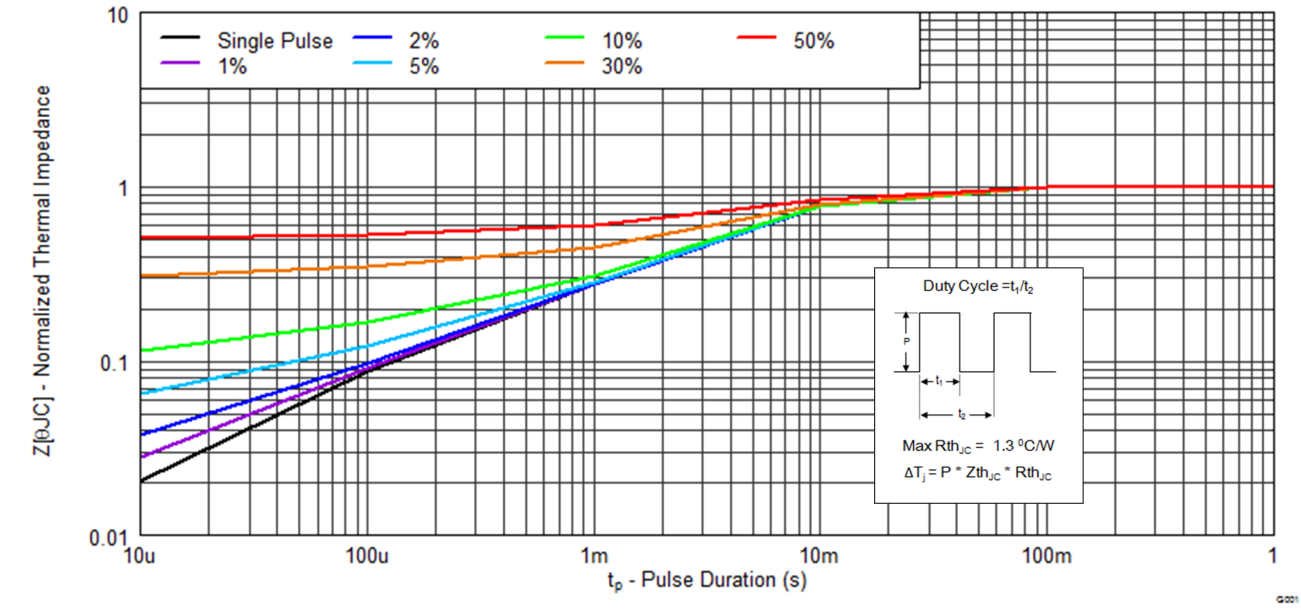


Figure 1. Transient Thermal Impedance

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Typical MOSFET Characteristics (continued)

(T_A = 25°C unless otherwise stated)

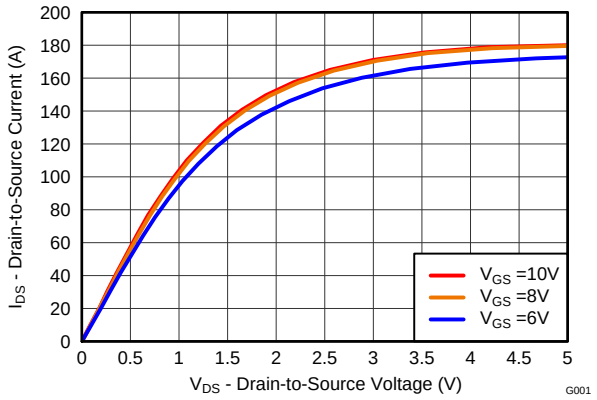


Figure 2. Saturation Characteristics

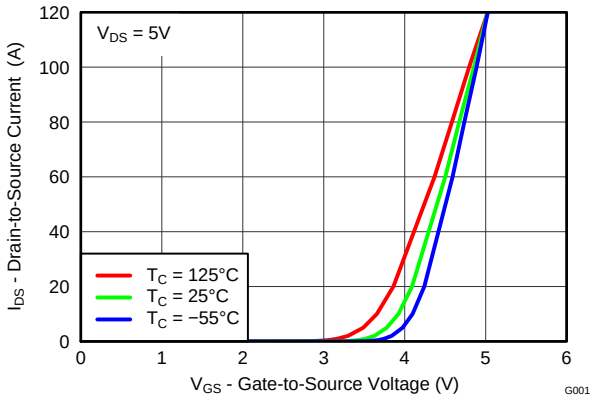


Figure 3. Transfer Characteristics

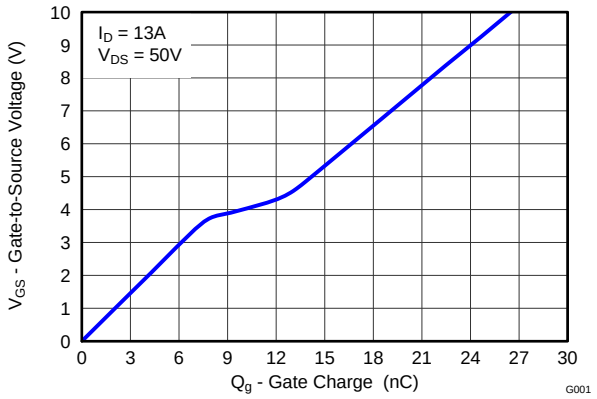


Figure 4. Gate Charge

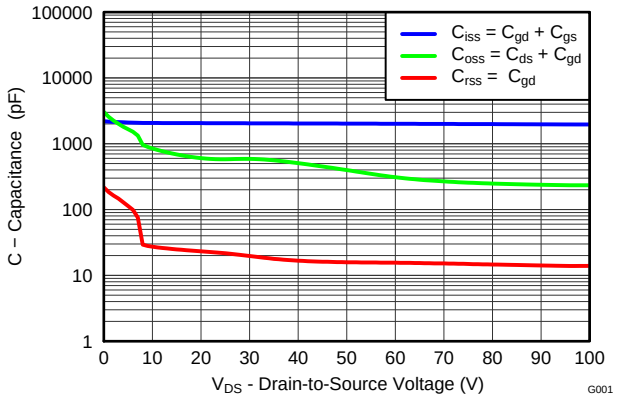


Figure 5. Capacitance

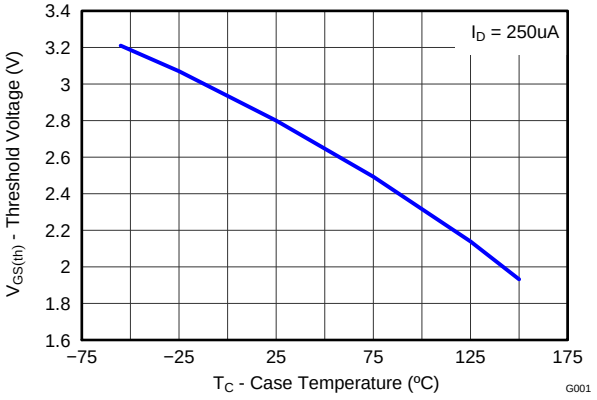


Figure 6. Threshold Voltage vs Temperature

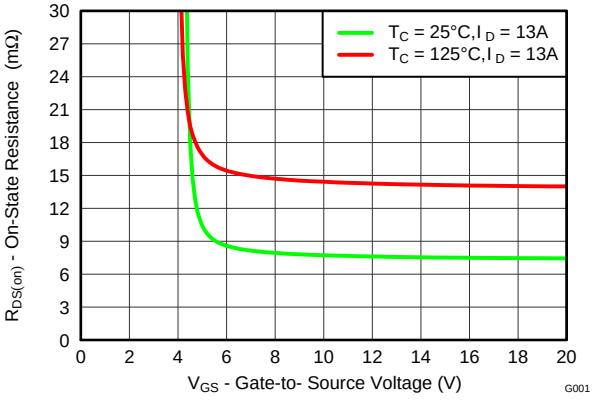


Figure 7. On-State Resistance vs Gate-to-Source Voltage

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Typical MOSFET Characteristics (continued)

(T_A = 25°C unless otherwise stated)

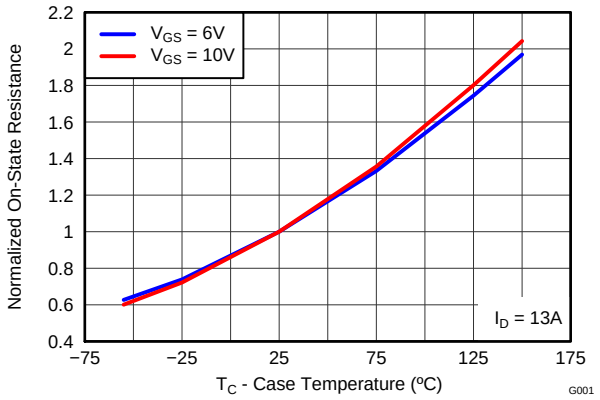


Figure 8. Normalized On-State Resistance vs Temperature

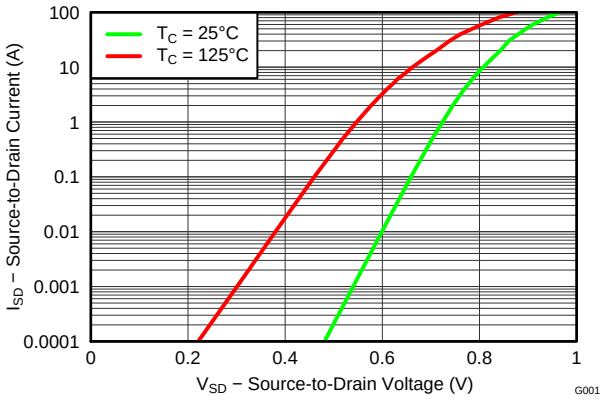


Figure 9. Typical Diode Forward Voltage

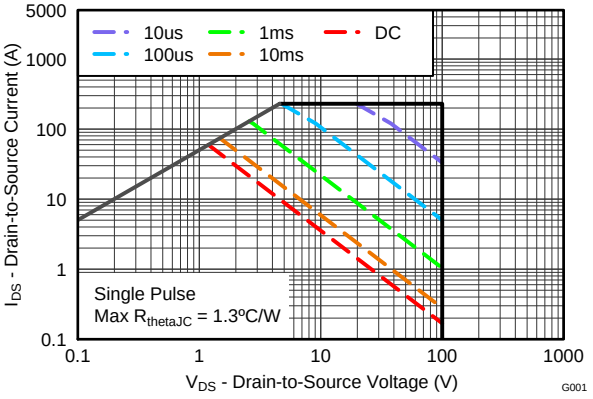


Figure 10. Maximum Safe Operating Area

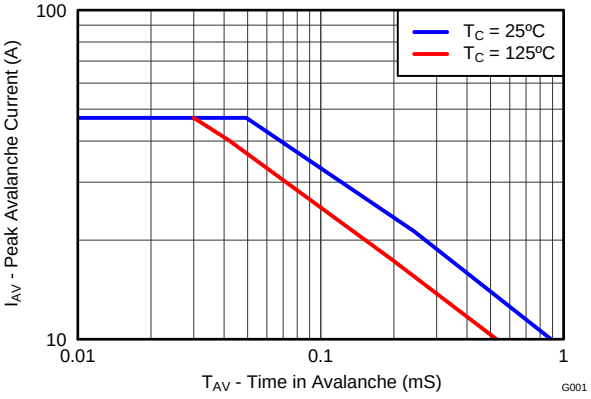


Figure 11. Single Pulse Unclamped Inductive Switching

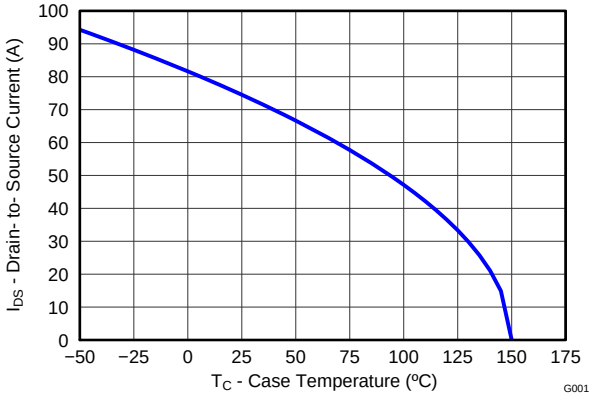


Figure 12. Maximum Drain Current vs Temperature

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6 Device and Documentation Support

6.1 Trademarks

NexFET is a trademark of Texas Instruments.

6.2 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

6.3 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms and definitions.

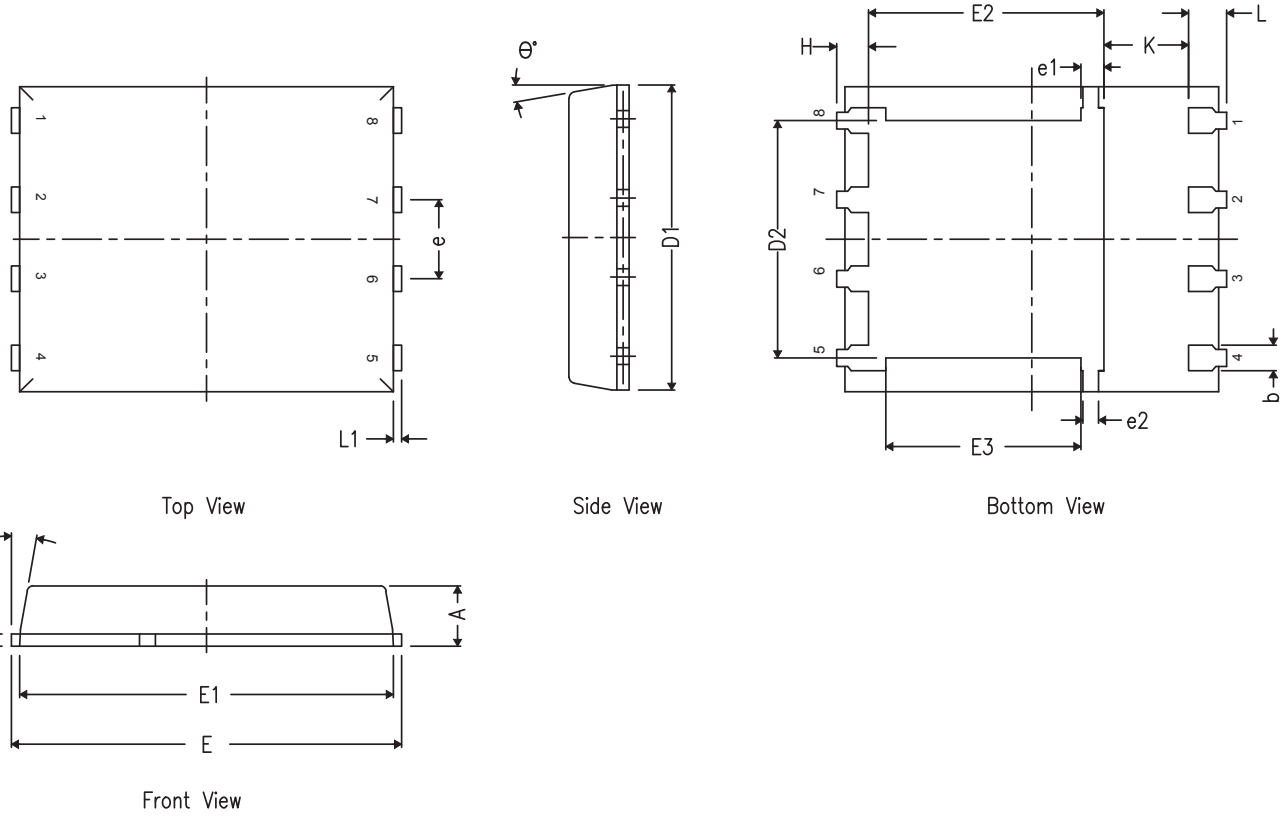
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7 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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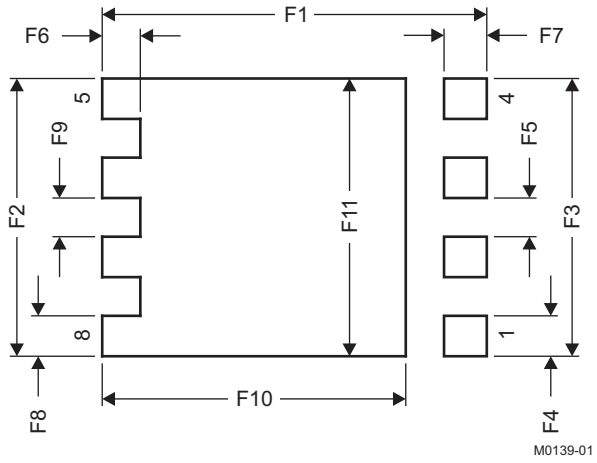
7.1 Q5A Package Dimensions



DIM	MILLIMETERS		
	MIN	NOM	MAX
A	0.90	1.00	1.10
b	0.33	0.41	0.51
c	0.20	0.25	0.34
D1	4.80	4.90	5.00
D2	3.61	3.81	4.02
E	5.90	6.00	6.10
E1	5.70	5.75	5.80
E2	3.38	3.58	3.78
E3	3.03	3.13	3.23
e	1.17	1.27	1.37
e1	0.27	0.37	0.47
e2	0.15	0.25	0.35
H	0.41	0.56	0.71
K	1.10	–	–
L	0.51	0.61	0.71
L1	0.06	0.13	0.20
θ	0°	–	12°

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7.2 Recommended PCB Pattern

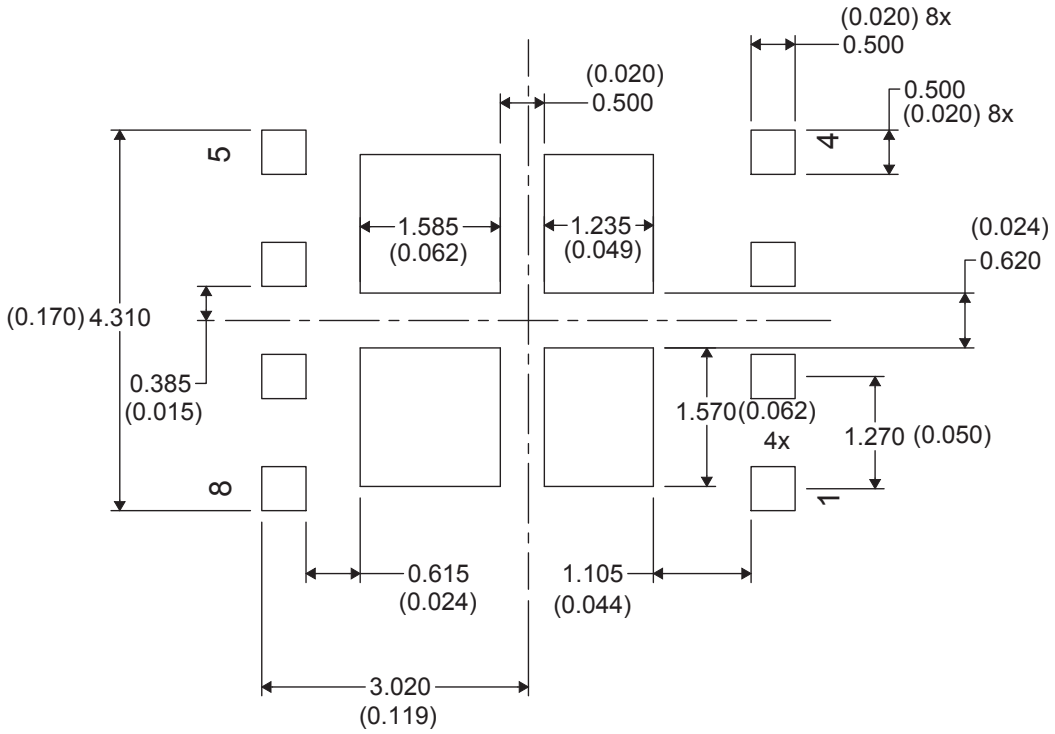


DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
F1	6.205	6.305	0.244	0.248
F2	4.46	4.56	0.176	0.18
F3	4.46	4.56	0.176	0.18
F4	0.65	0.7	0.026	0.028
F5	0.62	0.67	0.024	0.026
F6	0.63	0.68	0.025	0.027
F7	0.7	0.8	0.028	0.031
F8	0.65	0.7	0.026	0.028
F9	0.62	0.67	0.024	0.026
F10	4.9	5	0.193	0.197
F11	4.46	4.56	0.176	0.18

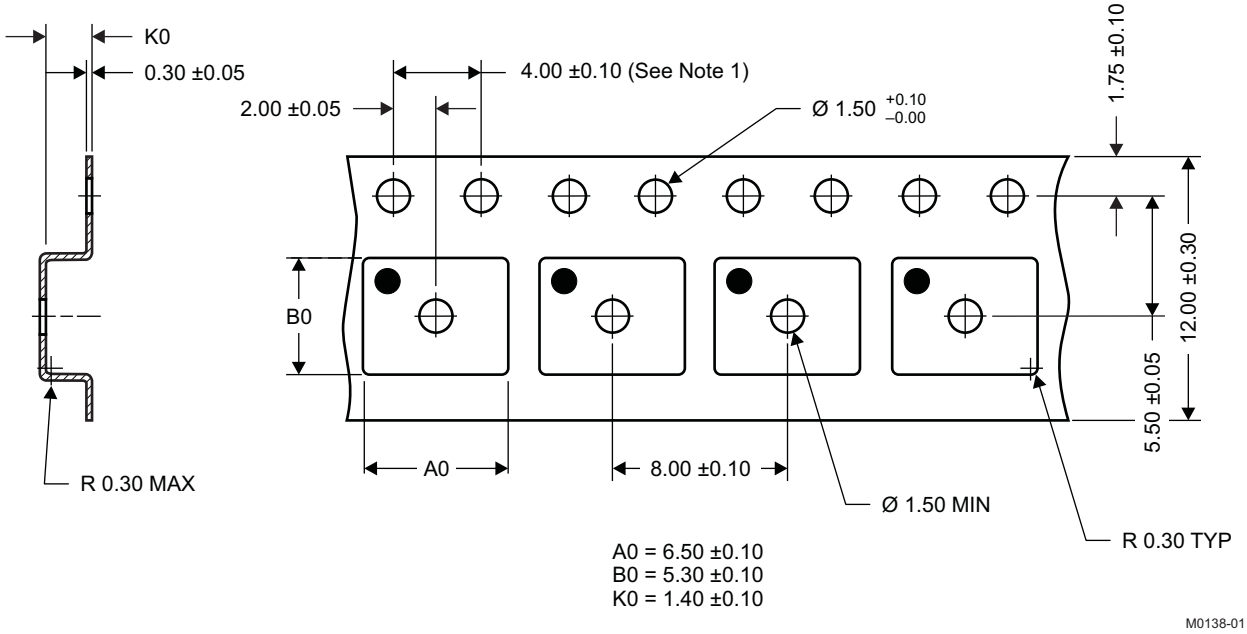
For recommended circuit layout for PCB designs, see application note SLPA005 – *Reducing Ringing Through PCB Layout Techniques*.

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7.3 Recommended Stencil Opening



7.4 Q5A Tape and Reel Information



Notes:

1. 10-sprocket hole-pitch cumulative tolerance ±0.2
2. Camber not to exceed 1 mm in 100 mm, noncumulative over 250 mm
3. Material: black static-dissipative polystyrene
4. All dimensions are in mm (unless otherwise specified)
5. A0 and B0 measured on a plane 0.3 mm above the bottom of the pocket

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CSD19533Q5A	Active	Production	VSONP (DQJ) | 8	2500 | LARGE T&R	ROHS Exempt	SN	Level-1-260C-UNLIM	-55 to 150	CSD19533
CSD19533Q5A.B	Active	Production	VSONP (DQJ) | 8	2500 | LARGE T&R	ROHS Exempt	SN	Level-1-260C-UNLIM	-55 to 150	CSD19533
CSD19533Q5AT	Active	Production	VSONP (DQJ) | 8	250 | SMALL T&R	ROHS Exempt	SN	Level-1-260C-UNLIM	-55 to 150	CSD19533
CSD19533Q5AT.B	Active	Production	VSONP (DQJ) | 8	250 | SMALL T&R	ROHS Exempt	SN	Level-1-260C-UNLIM	-55 to 150	CSD19533

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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Last updated 10/2025